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(71) Applicant: Zhang Xueliang

Address: Wenming Village Group,
Shuangqiao Town, Qidong County,
Hengyang City, Hunan Province,
421600

(72) Inventor Zhang Xueliang

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(54) Invention Title

Variable magnetic pole logarithmic rotating
magnetic field stepless speed change
electromagnetic transmission device

(57) Abstract

A variable magnetic pole logarithmic rotating magnetic field stepless speed change electromagnetic transmission device, the principle of which is derived from a three-phase generator and a three-phase asynchronous motor. The device is equivalent to the electric energy generated by the three-phase generator acting entirely on the rotor of the asynchronous motor, turning it into kinetic energy of the rotor and thermal energy generated by the rotor resistance. The device consists of two rotors, one of which is equipped with multiple excitation windings, each of which has an excitation voltage and direction controlled by computer numerical control to generate one or more pairs of rotating magnetic fields as needed; the other rotor is equipped with multiple guide bars arranged in a regular pattern, with both ends of each guide bar connected to a short-circuit ring. There is an air gap between the two rotors.

1 . A variable magnetic pole logarithmic rotating magnetic field stepless speed change electromagnetic transmission device, which combines the functions of a gearbox and a mechanical clutch. The device consists of two parts: an inner rotor and an outer rotor. One part is a squirrel cage rotor, while the other part is a rotor equipped with excitation windings to generate a rotating magnetic field, and vice versa.

2 . On the basis of meeting the above conditions, the rotor equipped with excitation windings to generate a rotating magnetic field, each individual excitation winding on it is controlled by a computer numerically to control the magnitude and direction of the excitation voltage, so as to generate one or more pairs of rotating magnetic fields according to actual needs.

Variable magnetic pole logarithmic rotating magnetic field stepless speed change electromagnetic transmission device

technical field

[0001] The present invention relates generally to a method and apparatus for transmitting power between an engine and a load, and more particularly to an electromagnetic variable speed transmission, and particularly to a variable speed transmission for a vehicle, for transmitting torque and power from a shaft rotating at one speed to a shaft rotating at a different speed.

Background Technology

[0002] I. Operating Principle and Function of Clutch in Transportation Vehicles: 1. Basic Principle

Working principle of clutch Working principle of clutch

The clutch, as the name suggests, uses "off" and "on" to transmit the appropriate amount of power. The clutch consists of friction plates, spring plates, pressure plates, and power output shafts, arranged between the engine and the transmission, to transmit the torque stored on the engine flywheel to the transmission, ensuring that the vehicle transmits the appropriate amount of driving force and torque to the drive wheels under different driving conditions. It belongs to the category of powertrain. During semi-engagement, the power input end and power output end of the clutch allow a speed difference, which is used to transmit the appropriate amount of power through its speed difference.

[0003] 2. Function:

Through the correct manipulation of the driver, the engagement and disengagement of the front and rear parts of the clutch (engine and transmission) can be achieved. It can indirectly achieve the following functions: (1) starting; (2) shifting; (3) reversing; (4) braking.

[0004] II. Currently, the gearboxes used in the automotive transportation sector primarily fall into the following categories:

1. Broadly classified based on their operational principles, they can be divided into three types:

step-variable transmission

It is currently the most widely used type. It uses gear transmission and has several fixed transmission ratios. Depending on the type of gear train used, there are two types: fixed-axis transmission (common transmission) and rotating-axis transmission (planetary gear transmission). Currently, the transmission ratio of passenger cars and light and medium-sized trucks usually has 3-5 forward gears and one reverse gear. In the combined transmission used in heavy-duty trucks, there are more gears. The number of gears refers to the number of forward gears.

continuously variable transmission

Its transmission ratio can be infinitely varied within a certain range of values, and there are two common types: electric and hydraulic (dynamic fluid). The transmission component of the electric continuously variable transmission is a DC series motor, which is widely used in the transmission system of super heavy-duty dump trucks, in addition to its application in trolleybuses. The transmission component of the hydraulic continuously variable transmission is a hydraulic torque converter.

compound transmission

It refers to the hydraulic mechanical transmission composed of a hydraulic torque converter and a gear-type step-variable transmission, whose transmission ratio can be infinitely varied within several discontinuous ranges between the maximum and minimum values. It is currently widely used.

[0005] 2. According to the operation mode, it can be divided into the following three types: forced operation

type transmission

It is to rely on the driver to directly manipulate the shift lever to shift gears. Automatic control type transmission

The selection and shifting of the transmission ratio are automatically performed. The term "automatic" refers to the fact that the change of each gear position of the mechanical transmission is achieved by controlling the actuating elements of the shifting system with a signal system that reflects the engine load and vehicle speed. The driver only needs to manipulate the accelerator pedal to control the vehicle speed.

semi-automatic transmission

There are two types: one is the commonly used automatic operation of several gears, while the remaining gears are operated by the driver; the other is the pre-selected type, where the driver selects a gear in advance using buttons, and when the clutch pedal is pressed or the accelerator pedal is released, an electromagnetic device or hydraulic device is activated to shift gears.

[0006] 3. Advantages and disadvantages of various types of gearboxes:

Manual transmission (MT): Its primary characteristic is high transmission efficiency (typically exceeding 80%). However, the overall performance of a manual transmission largely depends on the skill level of the driver. Skilled drivers can operate it smoothly and efficiently, saving fuel.

[0007] Automatic transmission AT: The structure is more complex, with the main components being a torque converter and a planetary gear set. The transmission efficiency is limited by the quality of the torque converter. Currently, automatic transmissions with a locking mechanism or even a multi-plate clutch structure (such as Mercedes-Benz's MCT) can significantly improve transmission efficiency.

[0008] Semi-automatic transmission A/MT: It is equivalent to installing a driver simulation shift mechanism on a manual transmission. The main advantages are low cost and high transmission efficiency (similar to a manual transmission). The disadvantages are general shift smoothness and low intelligence.

[0009] Continuously Variable Transmission (CVT): As the name suggests, a CVT has an infinite number of gears. Its biggest advantage is its smooth acceleration, with no concept of gear-shifting jolts. It offers excellent comfort. However, its disadvantage is that it can only withstand a limited amount of torque. If the engine power is too strong, the transmission is prone to slipping and overheating. Although CVTs are less efficient than manual transmissions, they can maximize the engine's operation at its most efficient speed, thus achieving better fuel economy.

[0010] Dual-clutch transmission (DCT): It is divided into two designs: dry and wet clutches. The dry type can withstand a smaller torque range and is suitable for small-displacement turbocharged engines. The wet structure has better heat dissipation, but the transmission efficiency is lower than the dry type. The biggest advantage of dual-clutch is that it uses two sets of clutches to control gear shifting. When one gear is disconnected, it means that the adjacent gear is engaged, so the gear shift is almost instantaneous and very fast. If the problems of clutch plate heat generation and heat dissipation can be better solved, dual-clutch is a very suitable choice for sports cars.

Invention content

[0011] The present invention relates to an electromagnetic continuously variable transmission device for transmitting kinetic energy between a pair of independent rotating shafts. This electromagnetic continuously variable transmission device includes a hollow cylindrical outer rotor part and an inner rotor part. The inner rotor part is disposed in the central hole of the outer rotor part and can independently rotate within the central hole of the outer rotor. The outer rotor part can independently rotate around the inner rotor part in the circumferential direction. There is an air gap between the outer rotor and the inner rotor. A pair of excitation coils are disposed on the inside of the outer rotor and face this air gap. The inner rotor part is similar to the rotor winding of a three-phase squirrel-cage induction motor, and its structure is embedded in a wire slot with copper bars (or other high conductivity metals) as conductors. The two ends of the copper bars (or other high conductivity metals) are welded together with shorting rings. The outer rotor part and the inner rotor part can simultaneously rotate in one direction. In response to the rotation of the outer rotor part and the inner rotor part, magnetic flux paths are generated between multiple pairs of excitation coils, air gaps, and squirrel-cage rotor core parts, forming a rotating

magnetic field. The magnetic flux paths induce electrical power in the squirrel-cage windings, and under the interaction of electromagnetic forces, mechanical power is transmitted between the outer rotor and the inner rotor.

[0012] The idea of this invention comes from the working principle of three-phase asynchronous motor.

When a symmetrical current is passed through the three-phase stator winding

When three-phase AC power is applied, a rotating magnetic field is generated that rotates clockwise along the stator and rotor inner circle space at a synchronous speed n_1

Rotating magnetic field. As the rotating magnetic field rotates at a speed of n_1 , the rotor conductor, which is initially stationary, will cut the stator rotating magnetic field and generate induced electromotive force (the direction of the induced electromotive force is determined using the right-hand rule). Since the two ends of the rotor conductor are short-circuited by a short-circuit ring, an induced current that is basically consistent with the direction of the induced electromotive force will be generated in the rotor conductor under the action of the induced electromotive force. The current-carrying conductor of the rotor is subjected to electromagnetic force in the stator magnetic field (the direction of the force is determined using the left-hand rule). The electromagnetic force produces electromagnetic torque on the rotor shaft, driving the rotor to rotate in the direction of the rotating magnetic field. When three-phase symmetrical AC current is passed through the three-phase stator windings (each phase is 120 degrees apart in electrical angle), a rotating magnetic field will be generated, which cuts the rotor windings, thereby generating induced current in the rotor windings (the rotor windings are a closed circuit). The current-carrying rotor conductor will generate electromagnetic force under the action of the stator rotating magnetic field, forming electromagnetic torque on the motor shaft, driving the motor to rotate, and the direction of rotation of the motor is the same as the direction of the rotating magnetic field.

[0013] We know that the reason why a three-phase motor can rotate is under the action of a rotating magnetic field. The rotating magnetic field of a three-phase motor is generated by the interaction of three-phase current flowing through the stator windings

The main characteristics of the field are as follows:

The magnetic field in the air gap of an AC motor is named after its continuous rotation along the cylindrical surface of the stator and rotor cores. The rotating magnetic field is the basic condition for the mutual conversion between electrical energy and rotational mechanical energy.

[0014] We know that when the engine is working, the engine drives the outer rotor part to rotate at high speed. We only need to load a magnetic field on the outer rotor to form a rotating magnetic field. We can control the size and direction of the excitation voltage loaded on each physical excitation winding of the rotor through a computer numerical control circuit, and generate one or more pairs of rotating magnetic fields according to needs. This is equivalent to changing the speed of the rotating magnetic field, thus changing the speed and output torque of the squirrel cage rotor, achieving the purpose of variable speed and variable torque.

[0015] Advantage of this invention is the actual torque transmission between two simultaneously rotating rotors, which is achieved through rotating magnetic fields without any mechanical transmission connection between the inner and outer rotors. Through electromagnetic induction and the interaction of electromagnetic fields, torque and power are transmitted between the inner and outer rotors. Since there is no mechanical connection, smooth and rigid operation can be achieved. [0016] Another advantage of this invention is the use of squirrel cage rotors, which improves energy transmission efficiency and operational reliability, with the advantages of simple structure, low failure rate, and simple maintenance. The energy transmission efficiency of asynchronous motors is comparable to gearboxes, and compared to gearboxes, speed conversion is smoother and faster, with no stuttering sensation during speed conversion. Mechanical and hydraulic belt-type continuously variable transmissions are prone to belt slippage, low energy transmission efficiency, and are not suitable for high-power transmission. This device does not exhibit the above phenomena.

[0017] Another advantage of the present invention is that it can easily achieve integrated control of numerical control automation, making the device easy to promote and apply.

[0018] Another advantage of the present invention is its high integration, which can completely replace the clutch and gearbox in vehicles. When the outer rotor has zero excitation current, the inner rotor has no power output, achieving power isolation between the inner and outer rotors. This can serve as a clutch in transportation vehicles, reducing the need for a clutch component and thus reducing manufacturing costs and mechanical failure rates.

Illustration of the drawings

[0019] Figure 1 is a typical structural diagram of the invention, which forms a

rotating magnetic field with a pair of magnetic poles.

- [0020] Figure 2 shows a rotating magnetic field with two pairs of magnetic poles.
- [0021] Figure 3 shows a rotating magnetic field with 3 pairs of magnetic poles.
- [0022] Figure 4 shows a rotating magnetic field with 4 pairs of magnetic poles.
- [0023] Figure 5 shows a rotating magnetic field with 6 pairs of magnetic poles.
- [0024] Figure 6 shows a rotating magnetic field with 12 pairs of magnetic poles.

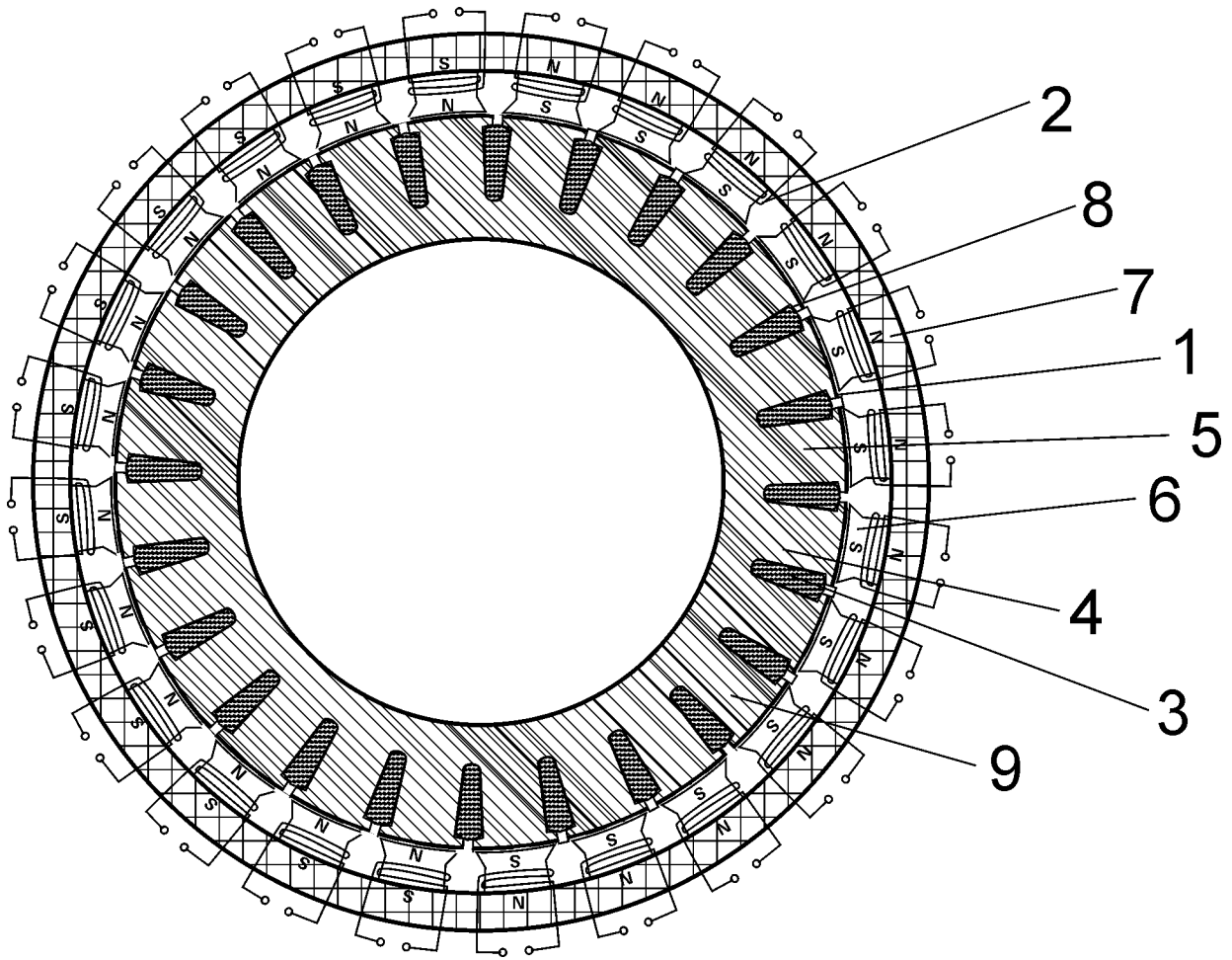


Figure 1

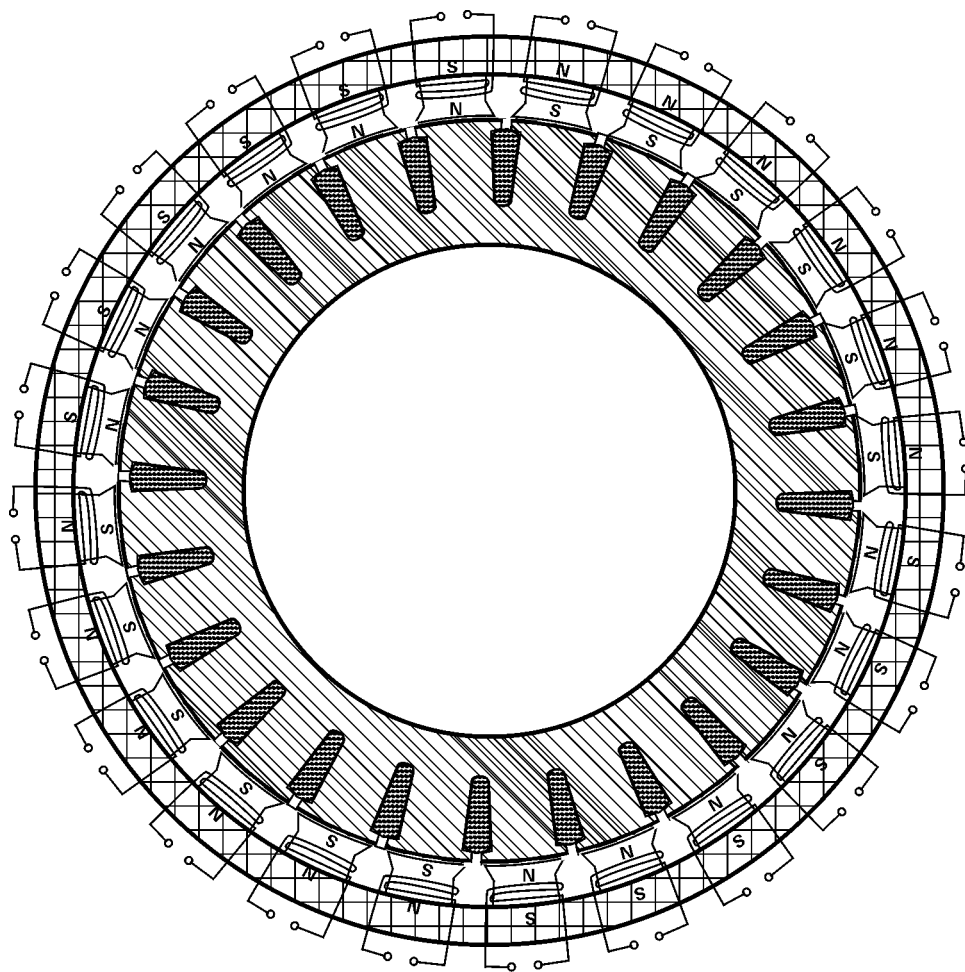


Figure 2

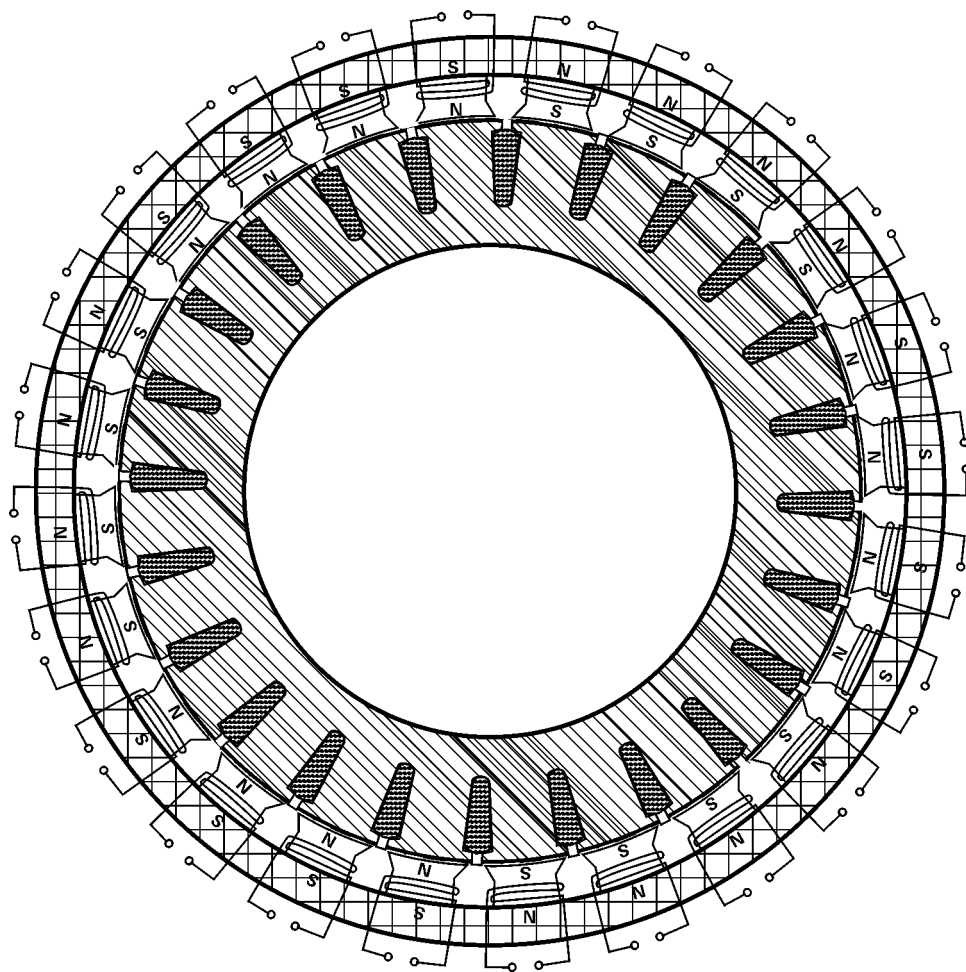


Figure 3

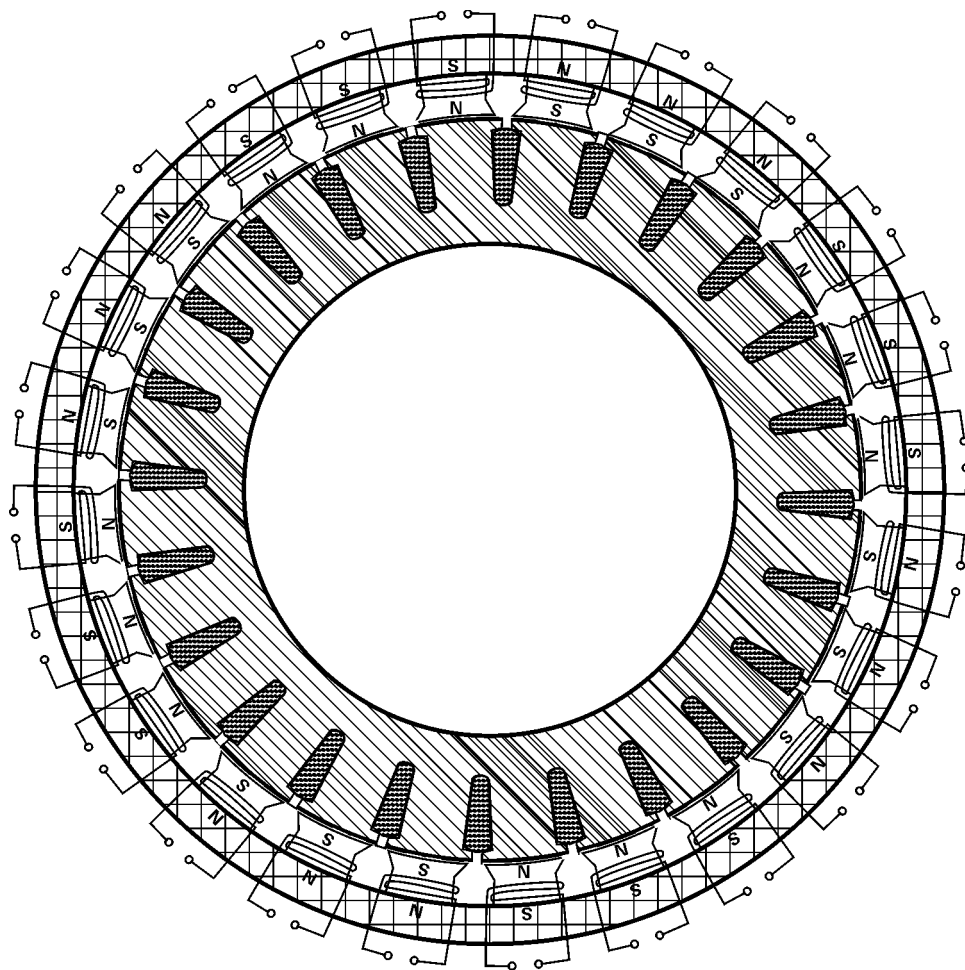


Figure 4

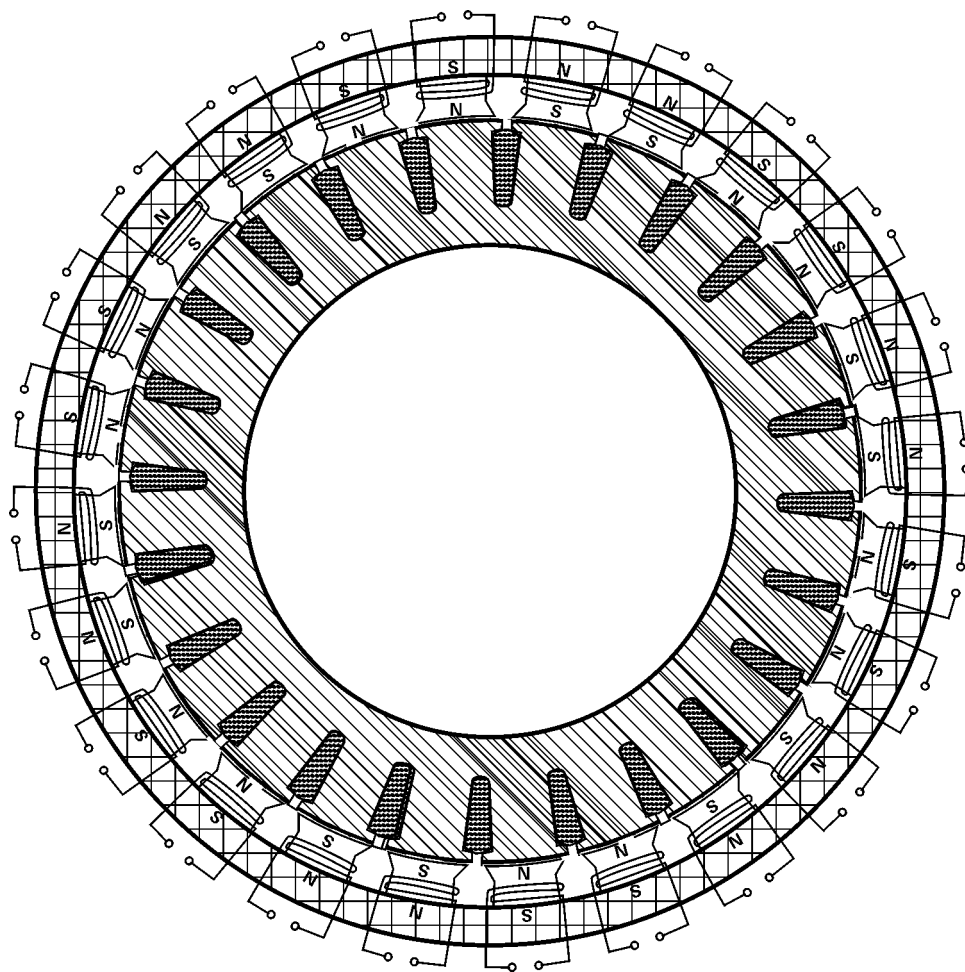


Figure 5

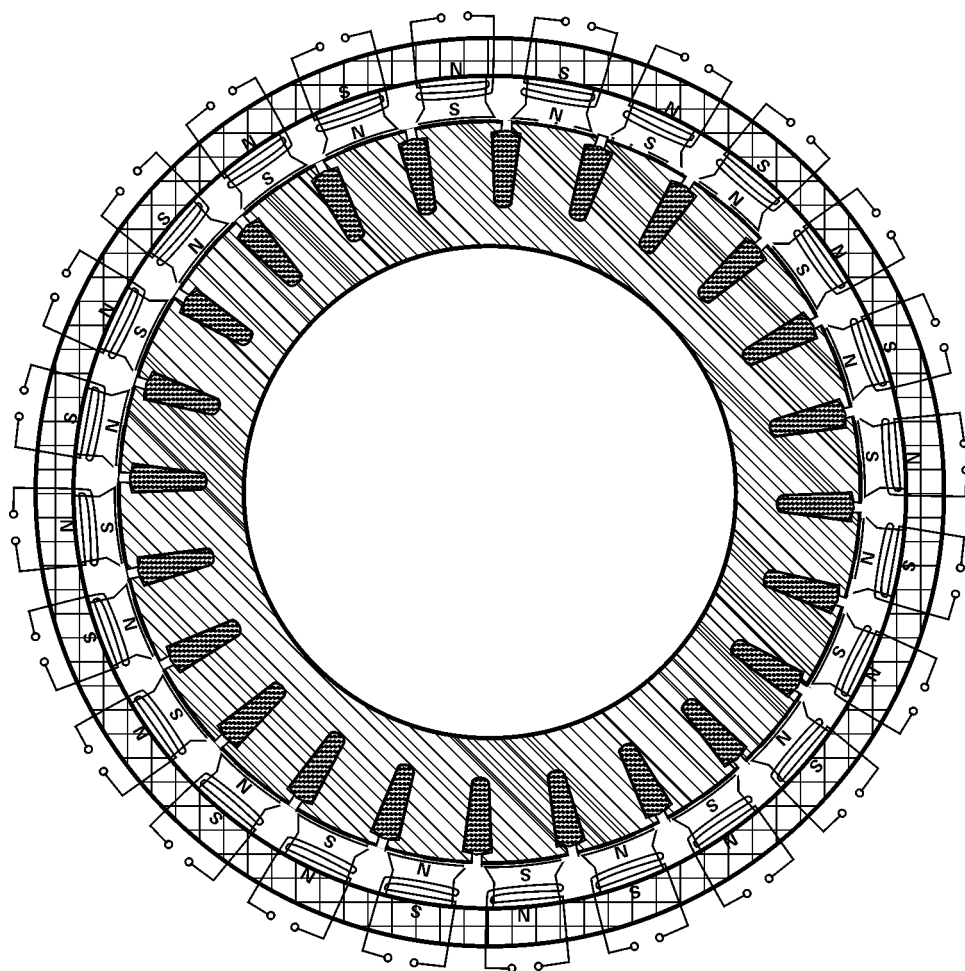


Figure 6